# Polestar Automotive Holding UK PLC (PSNY)

| RECOMMENDATION | SELL |
|---|---|
| Valuation Date | 21/04/2026 |
| Target Price ($) | 15,14 |
| Current price ($) | 18,970000 |
| 52 Week High | 42,60 |
| 52 Week Low | 11,75 |
| Upside (%) | 18,61 |
| MKTCAP (M $) | 1,755 |
| EV (M $) | 6,682 |

**Figure 1: Summary**




**Figure 2: Polestar 2**




**Figure 3: Polestar 3**




**Figure 4: Polestar 4**




**Figure 5: Polestar 5**

## RECOMMENDATION

We initiate coverage on Polestar with a **SELL** recommendation **[Figure 1]**. While the company benefits from strong premium positioning and support from Volvo and Geely, we believe its valuation does not reflect its structural financial weaknesses.

Polestar operates with a structurally unprofitable cost base, with COGS reaching 143% of revenue in 2024, reflecting a lack of scale and limited cost control. The company has also consistently destroyed value, with returns remaining below its cost of capital. In addition, its highly leveraged balance sheet and limited liquidity (~6.5 months of runway) make it dependent on continued external financing.

Although a recovery scenario exists, it requires the simultaneous execution of multiple uncertain drivers, while downside risks are already materializing. Overall, we believe the risk-reward profile is unfavorable and see no clear path to sustainable profitability.

## INTRODUCTION

**Polestar** (Nasdaq: PSNY) is the Swedish premium electric performance car brand with a focus on uncompromised design and innovation, and the ambition to accelerate the change towards a sustainable future. Originally a Swedish racing team founded in 1996, Polestar evolved into Volvo's performance division before being relaunched as an independent, premium 100% electric brand. Now majority-owned by Volvo Cars and Geely, Polestar operates as a publicly traded company (entering the market via a SPAC merger at a $20 billion valuation). This unique partnership provides the company with vital access to global manufacturing plants and R&D resources that typical startups lack. **[1]**

### Main activity [Figure 2-6]

Polestar has four models in its line-up: Polestar 2 (fastback), Polestar 3 (luxury SUV), Polestar 4 (SUV coupé), and Polestar 5 (luxury GT). Planned models include Polestar 7 (compact SUV) to be introduced in 2028 and the Polestar 6 (roadster). With its vehicles currently manufactured on two continents, North America and Asia, Polestar is diversifying its manufacturing footprint further, with production of Polestar 7 planned in Europe. **[1]**

It also operates "Polestar Energy," a smart ecosystem that essentially turns a car into a power bank for a home and the grid. Using a dedicated app, it automatically charges the vehicle when electricity is cheapest and greenest, while its bi-directional technology allows it to use the car's battery to power the house during blackouts or even sell energy back to the grid for a profit. **[2]**

| Model | Price | Architecture | Fast Charging | Positioning |
|---|---|---|---|---|
| Polestar 2 | €55k-60k | 400V | 180 kW (22-28 min 10-80%) | Accessible premium |
| Polestar 3 | $68k | 400V | 250 kW (~25 min) | Premium SUV |
| Polestar 4 | $55k | 400V | 200 kW (~26 min) | SUV coupé |
| Polestar 5 | $125k | 800V | 350 kW (~10 min 10-80%) | Performance flagship |

**Figure 6: Polestar Market Segmentation**




**Figure 7: Polestar Market Expansion**

| Region | Sales Share | 2026 Sales Points Target | Local Production |
|---|---|---|---|
| Europe | 60% | ~130 (+75% vs 2024) | Slovakia (2028) |
| North America | 25% | ~60 | Charleston, USA |
| Asia-Pacific | 15% | Expanding | China, South Korea |

**Figure 8: Polestar Sales Target**




**Figure 9: Waste, Polestar operation**

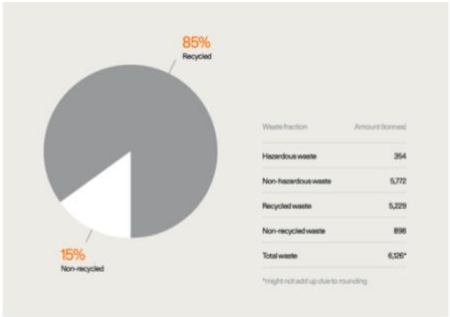



**Figure 10: Waste, manufacturing**

## Segments & Geography [1]

Polestar operates in a unique segment (premium electric vehicles) with an upmarket strategy, shifting its product mix toward higher-margin models (Polestar 3/4/5) away from the entry-level Polestar 2. Headquartered in Gothenburg, Sweden, its cars are available in 28 markets globally across North America, Europe, and Asia Pacific **[Figure 7]**.

The asset-light infrastructure features R&D concentrated in Sweden (Gothenburg) and UK (Coventry) with ~2,500 global employees, while production is entirely outsourced across China, USA (Charleston), South Korea (Busan), and Slovakia (Košice from 2028) **[Figure 8]**.

## Red Line

Polestar's investment case ultimately hinges less on its ability to scale as an independent player and more on its capacity to remain a going concern in an increasingly competitive and capital-intensive EV market. Given its ongoing losses, negative equity position, and reliance on external financing, the key question is not only scaling toward profitability, but whether the company can sustain operations on a standalone basis or will need to rely on strategic support, restructuring, or a potential acquisition by a larger automotive or industrial group to survive.

# BUSINESS DESCRIPTION

## Business Model

Polestar operates an asset-light model, without owning manufacturing facilities and relying on strategic partnerships with Volvo Cars and Geely Group for global manufacturing across multiple regions: China, the US, South Korea, and Slovakia by 2028. This approach provides operational flexibility and reduces capital intensity. The value proposition is built on three inseparable pillars: distinctive Scandinavian minimalist design, high-performance electric platforms (up to 884 hp on the Polestar 5 with 800V technology), and radical sustainability commitment with 100% renewable electricity in production and full $CO_2$ transparency for each vehicle. Advanced electric platforms (400V and 800V) ensure competitive performance and range, while native Android Automotive integration with OTA updates continuously enhances user experience **[3]**.

A key differentiator is the use of blockchain technology: Circulor to trace critical materials (cobalt, lithium, nickel), guaranteeing a verifiable ethical supply chain **[9]**. Customer data collected through 192 sales points and 1,269 service points provides direct feedback to optimize products and experience, though this exploitation remains less systematic than Tesla's approach. However, this model also limits control over production costs and increases dependence on external partners, which contributes to Polestar's current profitability challenges.

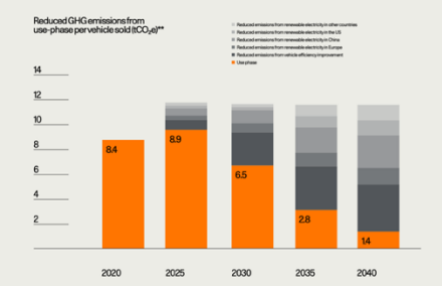



**Figure 11: GHG emissions**

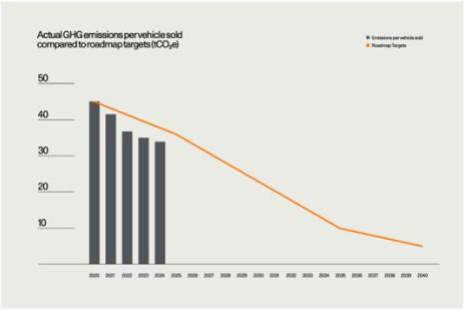



**Figure 12: GHG emissions vs Roadmap targets**

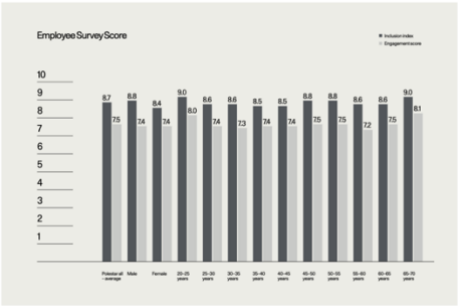



**Figure 13: Inclusion and engagement score**




**Figure 14: Average training hours per employee**

## Sources of income

Polestar's revenue model is evolving from near-total dependence on vehicle sales toward strategic diversification, with Q1-Q3 2025 revenues reaching $2,171M (+48.8%) driven by volume growth (44,482 vehicles, +36.5%) and premium product mix enrichment (Polestar 3 at $68k, Polestar 4 at $55k, Polestar 5 at $125k). The most significant development is the emergence of carbon credits as a strategic revenue source: $123M in 9M 2025 (vs $0.04M in 2024), representing approximately 6% of revenue, generated by selling excess $CO_2$ credits to ICE manufacturers constrained by EU, UK, and CARB regulations **[11]**.

## ESG OVERVIEW [4]

This sustainability positioning is a key component of Polestar's differentiation strategy and directly shapes its ESG profile. However, these strategic strengths must be assessed within the context of a rapidly evolving and highly competitive EV market.

## Environmental

Polestar has articulated a clear and ambitious environmental strategy, with targets to reduce greenhouse gas emissions per vehicle by 50% by 2030 (vs. 2020) and to achieve full climate neutrality across its value chain by 2040. These objectives are aligned with a 1.5°C pathway and embedded in a broader sustainability framework covering climate, transparency, circularity, and inclusion **[Figure 11-12]**.

A key differentiator is the Polestar 0 Project, which aims to deliver a fully climate-neutral vehicle by 2030 through the elimination of lifecycle emissions, rather than reliance on carbon offsetting.

Polestar recycles most of its waste in both operations and manufacturing, highlighting a strong overall waste management, especially considering the higher scale of manufacturing **[Figure 9-10]**.

## Social

Polestar follows a Code of Conduct aligned with international labor and human rights standards, ensuring fair employment, inclusion, and whistleblowing mechanisms. It prioritizes talent retention through training (7,000 hours in 2024) and maintains strong safety performance with zero major incidents **[Figures 13–14]**. Customer trust is reinforced by GDPR-compliant data governance.

## Governance [5]

Leadership changes, with Michael Lohscheller (CEO) and Jean-François Mady (CFO), strengthen turnaround and financial expertise. This is supported by an experienced executive team and a largely independent board. Governance is now focused on financial discipline, execution, and risk oversight, critical for addressing losses and delivering growth.

| | Polestar | Tesla | BYD |
|---|---|---|---|
| Market Cap (M $) | 1 754,80 | 1 472 829,55 | 133 777,95 |
| Enterprise Value (M $) | 6 930,54 | 1 437 874,55 | 156 236,04 |
| Price to Total Revenue per Share (FY0) | 1,09 | 16,73 | 1,11 |
| Price / Revenue (SmartEstimate ®) (FY1) | 0,48 | 14,66 | 1,05 |
| Price / Revenue (SmartEstimate ®) (FY2) | 0,28 | 12,42 | 0,97 |
| Enterprise Value to Total Revenue (FY0) | 3,19 | 15,40 | 1,25 |
| EV / Revenue (SmartEstimate ®) (FY1) | 1,82 | 14,31 | 1,21 |
| EV / Revenue (SmartEstimate ®) (FY2) | 1,08 | 12,13 | 1,11 |

**Figure 15: Comparative Financial Valuation**




**Figure 16: P/S Polestar and its competitors**




**Figure 17: EV/Revenue Polestar & Competitors**

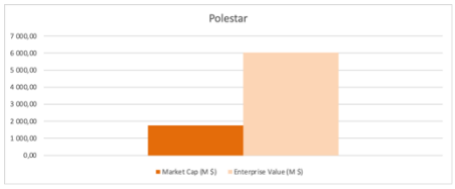



**Figure 18: Polestar Market Cap vs EV**

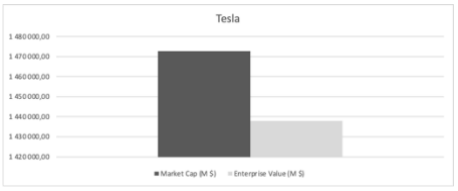



**Figure 19: Tesla Market Cap vs EV**

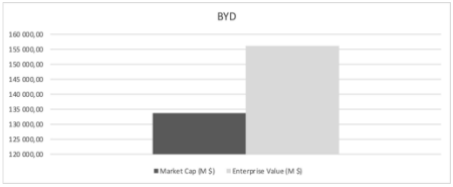



**Figure 20: BYD Market Cap vs EV**

## INDUSTRY OVERVIEW

The global electric vehicle (EV) market has expanded rapidly, with worldwide EV sales exceeding 20 million units in 2025 **[13]**, accounting for over 25% of new car sales. While Polestar has established itself as a premium, design-focused EV brand, it operates in a highly competitive and aggressive market.

### Competitors [6]

Polestar faces intense competition across the EV market. Tesla remains the dominant global player, with 1.64 million vehicles delivered in 2025 **[12]**, benefiting from significant scale advantages that translate into lower unit costs and stronger margins.

At the premium end, Porsche and traditional European manufacturers such as BMW and Mercedes-Benz compete on brand, performance, and customer loyalty. At the same time, Chinese players, particularly BYD, with 2.25 million EVs sold in 2025 **[10]**, are rapidly gaining market share through vertical integration and aggressive pricing strategies, increasingly targeting Europe.

As a result, Polestar is positioned between high-end legacy brands and lower-cost, high-volume competitors. This limits its pricing power, constrains volume growth, and delays the achievement of economies of scale, ultimately weighing on margins and profitability.

### Competitive Advantages

In this competitive environment, Polestar relies on differentiation rather than scale to position itself. Polestar follows a “differentiation strategy” meaning that it emphasizes transparency, sustainability, design, and software integration rather than seeking volume through price cuts, with 60,119 vehicles sold globally in 2025 (+34 % YoY), slightly outperforming the premium BEV segment. By leveraging an asset-light business model (utilizing Volvo/Geely manufacturing footprints) and industry-leading sustainability transparency (e.g., blockchain mineral tracing), it targets eco-conscious luxury consumers who prioritize high build quality and a future-proof tech stack. **[7]**

However, this differentiation strategy limits Polestar’s ability to compete on price and scale, reinforcing its dependence on premium positioning in an increasingly competitive market.

## PEER ANALYSIS [8, team analysis]

### Market Pricing

Competitive dynamics are directly reflected in Polestar's market valuation. As of April 2026, Polestar trades at a Price/Sales ratio of 1.09x - still at a deep discount to Tesla's 16.73x, yet now nearly on par with BYD's 1.11x **[Figure 16]**. This convergence with BYD is less a sign of recovery than a reflection of BYD's own valuation compression amid intensifying global

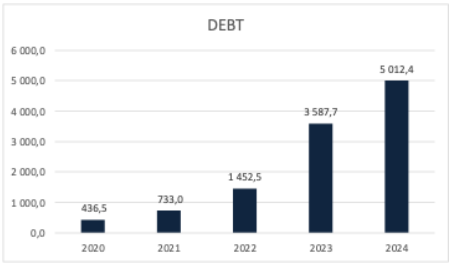



**Figure 21: Debt**

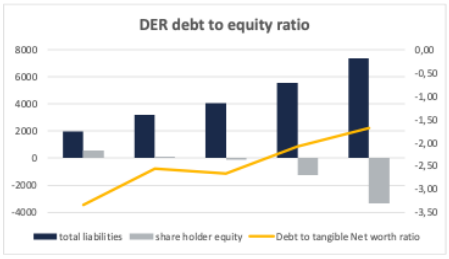



**Figure 22: Debt to Equity Ratio**

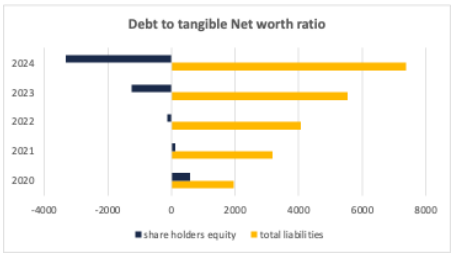



**Figure 23: Debt to tangible Net worth ratio**

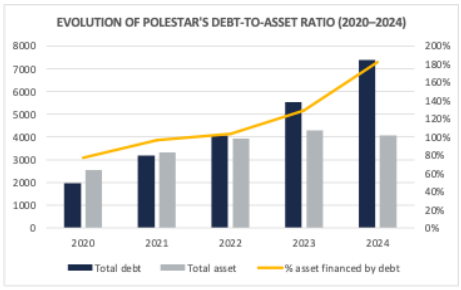



**Figure 24: Debt to Asset Ratio**

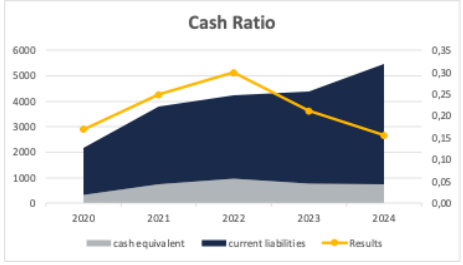



**Figure 25: Cash ratio**

competition and trade pressures. With a market cap of $1.75B, Polestar remains a small-cap player, dwarfed by Tesla's $1.47T and BYD's $134B **[Figure 15, 18-20]**. While Tesla continues to be valued as a high-growth technology platform and BYD as a high-volume manufacturing powerhouse, the market still prices Polestar as a capital-constrained challenger struggling to achieve meaningful scale. The absence of a P/E ratio and the absence of a positive EV/EBITDA further confirms that Polestar is not yet profitable at the operational level.

### Relative Valuation

The most telling metric in this dataset remains the relationship between Market Cap and Enterprise Value. Polestar's EV ($6.03B) is roughly 3.4x its Market Cap ($1.75B), confirming a capital structure heavily burdened by debt - consistent with its deteriorating balance sheet shown in the financial analysis **[Figure 18]**. In contrast, Tesla's EV ($1.44T) is slightly below its Market Cap, reflecting a large net-cash position and a fortress balance sheet **[Figure 19]**. Despite Polestar's modest P/S of 1.09x, its EV/Revenue stands at 3.19x - significantly above BYD's 1.25x and even above Tesla's EV/Revenue on a comparable basis **[Figure 17]**. This confirms that once debt obligations are factored in, Polestar is far from cheap: the equity market may appear to price it modestly, but its enterprise-level valuation reflects the full weight of its financing burden.

The valuation gap is fundamentally tied to business model maturity. Unlike BYD's high-volume, vertically integrated strategy that generates consistent cash flows, Polestar's premium-brand positioning has yet to reach the economies of scale necessary to justify its EV.

## FINANCIAL ANALYSIS & VALUATION

### Capital Structure

Polestar’s capital structure is a major constraint on its financial flexibility and valuation. Since 2020, total debt has increased from $437 million to approximately $5.0 billion **[Figure 21]**, while equity has deteriorated to around –$3.3 billion, reflecting accumulated losses. The company now operates with a highly leveraged balance sheet, with a debt-to-assets ratio exceeding 100% (approximately 123%), meaning liabilities exceed total assets. **[Figure 22]**

This structure implies that Polestar is primarily financed by creditors rather than shareholders, increasing refinancing risk. Interest expenses reached approximately $367 million in 2024, adding further pressure on an already loss-making model. **[Figure 23-24]**

Liquidity is also a critical constraint, with approximately $739 million in cash, corresponding to an estimated runway of 6.5 months without additional financing. Continued access to external capital is therefore required across all scenarios. **[Figure 25 - 26]**

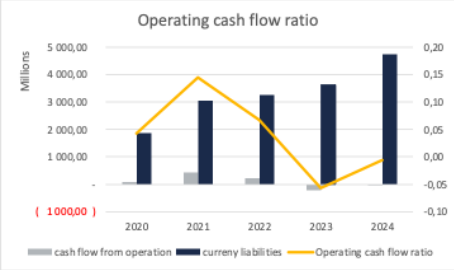



**Figure 26: Operating Cash Flow Ratio**

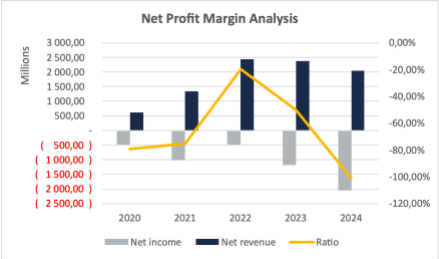



**Figure 27: Net Profit Margin Analysis**

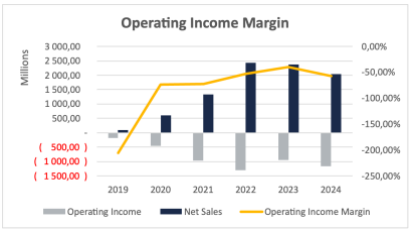



**Figure 28: Operating Income Margin**

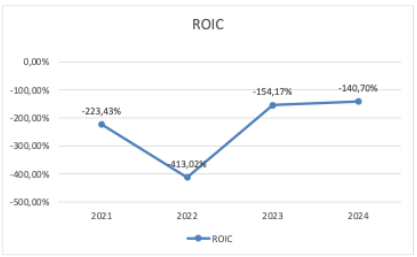



**Figure 29: ROIC**

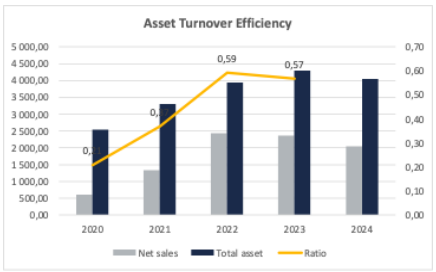



**Figure 30: Asset Turnover Efficiency**

In our model, we assume financing remains available in the Comeback and Grind scenarios, while limited access in the Fall scenario would significantly accelerate liquidity pressure.

## Financial Performance & Key Ratios

This financial structure is reflected in Polestar's operating performance and profitability metrics. Polestar's financial performance highlights a persistent inability to generate economic value. In 2024, the company reported a net margin of approximately –101% **[Figure 27]** and an operating margin of around –57% **[Figure 28]**, indicating that losses exceed revenues. Return on invested capital (ROIC) remains deeply negative at about –141% **[Figure 29]**, well below the cost of capital, confirming continuous value destruction.

From an efficiency perspective, asset turnover stands at approximately 0.49x **[Figure 30]**, below industry benchmarks, reflecting limited revenue generation relative to the asset base. Overall, these indicators suggest that Polestar's core issue lies in its cost structure rather than demand. To better understand these results, we break down the company's performance into its key operating drivers.

## Operating Drivers

### 1. Revenue

Polestar's revenue is driven by vehicle volumes and average selling prices (ASP), reflecting its premium positioning, as well as a growing contribution from carbon credits. The company experienced strong growth between 2019 and 2022, reaching $2.44 billion **[Figure 31]**, supported by the Polestar 2 ramp-up and favorable market conditions. However, this reversed in 2023 and 2024, with revenues declining to $2.03 billion, due to delayed launches, weaker premium EV demand in Europe, and increased competition from lower-cost manufacturers.

In our model, revenue projections depend on three factors: volume ramp-up of new models (Polestar 3, 4, 5), pricing dynamics, and overall EV market growth. In the Comeback scenario, revenue grows at approximately 20% annually, supported by successful execution and improved product mix. In the Grind scenario, growth remains moderate and gradually slows due to competitive pressure. In the Fall scenario, revenue stagnates and declines, reflecting a loss of market share.

Revenue growth is critical, as it drives economies of scale and margin improvement. Given the current cost structure, small changes in growth assumptions significantly impact valuation. However, this growth has not translated into improved cost efficiency, highlighting the company's inability to achieve economies of scale. **[Figure 32]**

### 2. COGS

Cost of Goods Sold is the most critical variable in Polestar's model, as the company operates with structurally negative gross margins. COGS fell

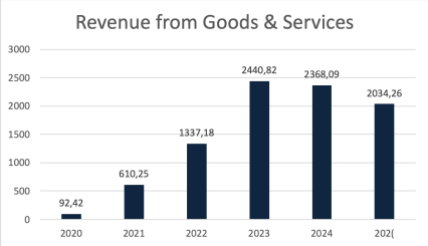



**Figure 31: Revenue Evolution**

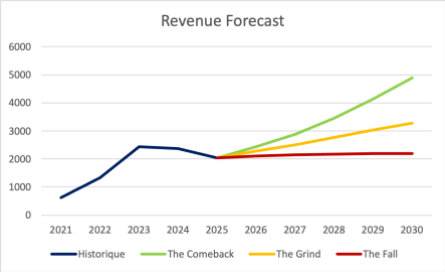



**Figure 32: Revenue Forecast**

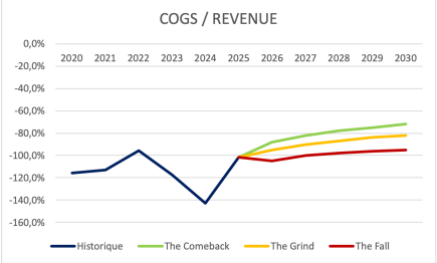



**Figure 33: Ratio COGS / Revenue**

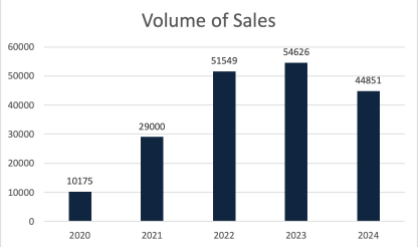



**Figure 34: Volume of Sales (data from the Annuel Report)**

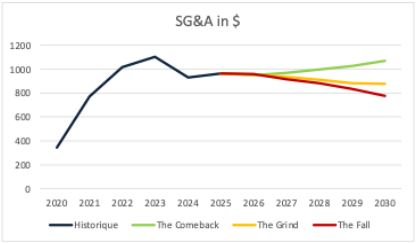



**Figure 35: SG&A in $**

below 100% of revenue only once, reaching 95.9% in 2022 under favorable conditions. It then increased to 117% in 2023 and 143% in 2024 **[Figure 33]**, meaning the company spends $1.43 for every dollar of revenue, generating a loss on each vehicle sold.

The main driver is a lack of scale. With production of around 45,000 vehicles per year **[Figure 34]**, Polestar cannot absorb fixed industrial costs linked to its outsourced model with Volvo and Geely. This creates a vicious circle: low volumes keep costs high, while high costs limit competitiveness and volume growth.

In our model, COGS is driven by scale effects, battery cost decline, and potential contract renegotiation. In the Comeback scenario, COGS declines rapidly to around 65% by 2035, enabling positive gross margins. In the Grind scenario, improvement is slower, reaching approximately 76%. In the Fall scenario, COGS remains above 100% for several years and only improves to around 91%, reflecting persistent inefficiencies. **[Figure 33]**

COGS is the most sensitive variable in the model, as small changes have a significant impact on profitability and valuation.

### 3. SG&A

SG&A is another major constraint, reaching $929 million in 2024, or 46% of revenue **[Figure 35]**. This reflects a fixed cost base designed for higher volumes, while the company operates at around 45,000 vehicles per year. As revenues declined, the SG&A ratio increased, reinforcing losses.

R&D spending also fell from $175 million in 2022 to $38 million in 2024 (1.9% of revenue), reflecting short-term cost control but raising concerns about long-term competitiveness.

In our model, SG&A is driven by operating leverage. In the Comeback scenario, strong growth reduces SG&A to around 13% of revenue. In the Grind scenario, it stabilizes at approximately 20%, while in the Fall scenario it remains elevated at around 32%, reflecting limited cost dilution. **[Figure 36]**

## Working Capital

Working capital is one of the few areas showing improvement. The cash conversion cycle decreased from 131 days in 2020 to 40 days in 2024 **[Figure 37]**, driven by better receivables collection and inventory management.

However, inventory remains a key risk, with around 135 days of inventory in 2024, indicating significant capital tied up in unsold vehicles. This makes working capital highly dependent on volume stability. **[Figure 38-40]**

In our model, we assume continued improvement in the Comeback scenario, stabilization in the Grind scenario, and deterioration in the Fall scenario, where lower volumes would reverse recent gains and increase

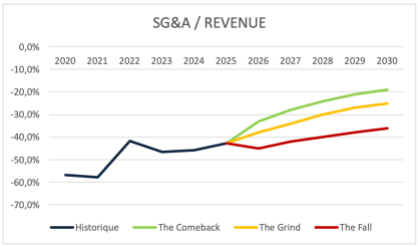



**Figure 36: SG&A/ Revenue**

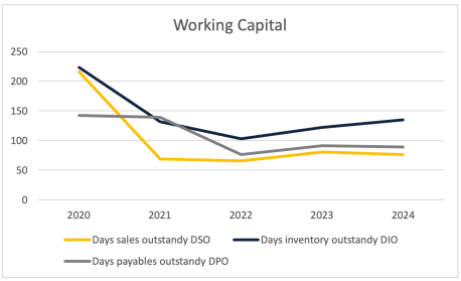



**Figure 37: Working Capital**

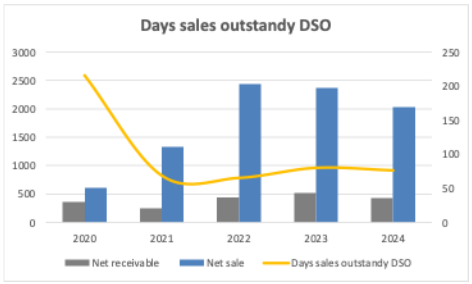



**Figure 38: DSO Ratio**

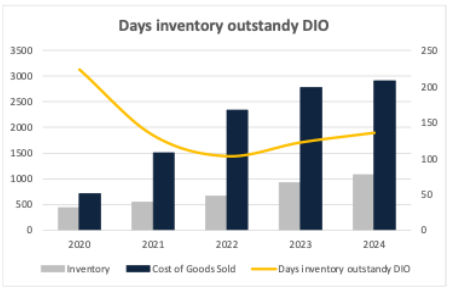



**Figure 39: DPO Ratio**

cash pressure. Compared to peers, Polestar remains less efficient, as leaders such as Tesla operate with a negative cash conversion cycle.

### Beta & WACC [Figure 41, Annex 1-2]

Our cost of capital is estimated using a CAPM-based asset approach. Given Polestar's deeply negative equity (−$3.33bn in 2024), we rely on the unlevered cost of capital (kA) rather than a traditional WACC. We use a risk-free rate of 4.0% and a market risk premium of 5–6%, capturing valuation uncertainty. The asset beta is set at 0.80, based on Damodaran's global automotive industry data, as more reliable than Polestar's volatile equity beta. This implies a cost of capital of ~8.0%, used as the discount rate. On the debt side, the cost of debt is around 7.3%, reflecting a ~330bps credit spread consistent with a B/B+ rating; we set βD = 0.26 accordingly. No tax shield is applied due to persistent losses. Overall, the implied WACC (kA) is ~8.0%, assuming a stable long-term operating structure.

## SCENARIO & SENSITIVITY ANALYSIS

### The 3 scenarios

Our three scenarios are derived from the evolution of key operating drivers, revenue growth, cost structure (COGS), operating expenses (SG&A), and working capital efficiency, which determine Polestar's ability to achieve scale and profitability.

**1. The Comeback [Figure 42]**

In the Comeback scenario, Polestar successfully executes its growth strategy, combining strong revenue expansion with significant cost improvements. Economies of scale progressively reduce COGS, while operating expenses are diluted as volumes increase. As a result, the company reaches profitability relatively early, and most of its valuation is driven by long-term stable cash flows. This leads to an enterprise value of approximately $13.6 billion, reflecting a successful transition toward a sustainable premium EV player.

**2. The Grind [Figure 43]**

In the Grind scenario, Polestar faces a more challenging environment. Growth remains moderate and cost efficiencies materialize only gradually, delaying the path to profitability. Competitive pressure limits margin expansion, and the company struggles to fully benefit from scale effects. Consequently, value creation is more limited, resulting in an enterprise value of around $11.6 billion.

**3. The Fall [Figure 44]**

In the Fall scenario, Polestar fails to achieve sufficient scale to offset its cost structure. High production costs persist, and ongoing losses prevent the company from generating positive cash flows. The firm becomes increasingly reliant on external financing, leading to value destruction over time. This is reflected in a negative enterprise value of approximately –

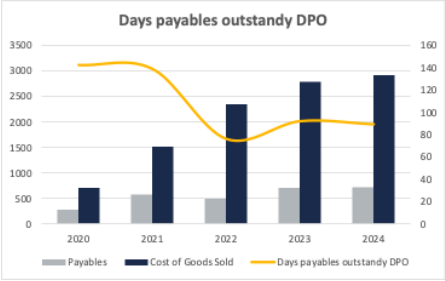



**Figure 39: DIO Ratio**

| Summary | Optimistic | Pessimistic | Model (base) |
|---|---|---|---|
| Risk-free rate (rf) | 4,00% | 4,00% | 4,00% |
| Market Risk Premium (MRP) | 5,00% | 6,00% | 5,50% |
| Asset Beta (βA) | 0,80 | 0,80 | 0,80 |
| Debt Beta (βD) | 0,26 | 0,26 | 0,26 |
| Cost of Debt kD (observed) | 7,31% | 7,31% | 7,31% |
| Cost of Equity kE | N/A | N/A | N/A |
| Unlevered Cost of Capital kA | 8,00% | 8,80% | 8,40% |
| WACC used in DCF | 8,00% | 8,00% | 8,00% |

**Figure 40: Beta & WACC**

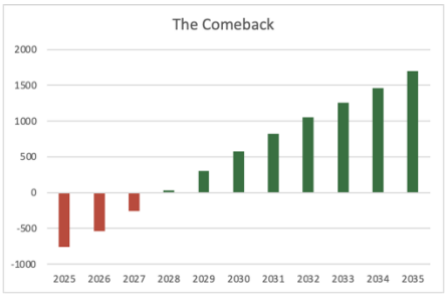



**Figure 42: NOPAT (M $) in the Best Case**

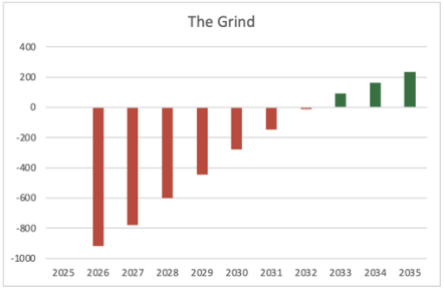



**Figure 43: NOPAT (M $) in Grind scenario**

$8.0 billion, highlighting the unsustainability of the business model under these assumptions.

## Sensitivity analysis

The valuation is highly sensitive to key assumptions, particularly WACC and terminal growth, reflecting the long-duration nature of Polestar's cash flows. A 1pp increase in WACC reduces enterprise value by approximately 14–17%, making it the most impactful variable in the model.

Terminal growth also plays a significant role, with a +0.5pp increase in g adding around 10% to valuation in the Comeback scenario, but only ~4% in the Grind scenario, highlighting the stronger convexity of value when long-term growth is achieved. **[Figure 45-47]**

Importantly, the Fall scenario yields consistently negative enterprise values, regardless of assumptions. This underscores the binary nature of the investment case: without reaching scale and profitability, valuation remains structurally negative. **[Figure 47]**

Overall, the dispersion across scenarios illustrates a high degree of valuation uncertainty, driven primarily by execution risk and the company's ability to scale sustainably.

## RISKS

We identify below the key risks that could further deteriorate our view, as well as the conditions under which we would revise our recommendation from Sell to Buy or to Hold **[Figure 48]**.

## Financing Risk

Polestar faces acute liquidity risk, as every capital raise either dilutes shareholders or deepens a $5B debt load. At –$3.3B equity and 182% debt-to-assets, access to capital markets is structurally uncertain, and a failed fundraising would accelerate the Fall scenario regardless of operational progress. We would revise our recommendation if Polestar secured a sizeable, non-dilutive financing facility such as a committed credit line from Geely or Volvo of at least $1.5B extending its runway beyond 18 months and reducing near-term refinancing risk materially.

## Geely/Volvo Dependency

Polestar doesn't control its factories, supply chain, nor its base technology. Volvo already cut its stake from 48% to 18% in 2024, explicitly citing Polestar as a resource drain, and escalating EU-China trade tensions raise the risk of adverse renegotiation of manufacturing terms. We would revise our position if Polestar formalized long-term supply and manufacturing agreements with Geely at fixed terms or diversified its production base sufficiently to reduce single-counterparty exposure.

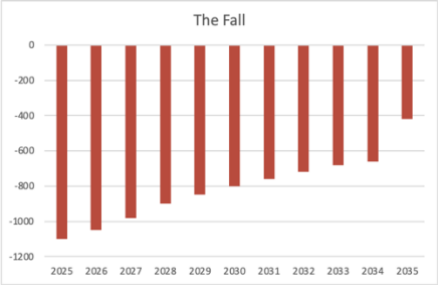



**Figure 44: NOPAT (M $) in the Worst Case**

| Sensitivity Analysis | | | | | | |
|---|---|---|---|---|---|---|
| WACC | Growth Rate (g) | | | | | Sensitivity to Base Case |
| 13 637,95 | 2,5 % | 2,8 % | 3,0 % | 3,3 % | 3,5 % | |
| 7,0 % | 16 456,70 | 17 383,77 | 18 426,74 | 19 608,76 | 20 959,64 | +35.1% |
| 7,5 % | 14 244,81 | 14 960,68 | 15 756,09 | 16 645,07 | 17 645,18 | +15.5% |
| 8,0 % | 12 452,03 | 13 016,75 | 13 637,95 | 14 324,53 | 15 087,40 | 0.0% |
| 8,5 % | 10 972,98 | 11 426,38 | 11 921,00 | 12 462,73 | 13 058,63 | -12.6% |
| 9,0 % | 9 734,74 | 10 104,17 | 10 504,40 | 10 939,42 | 11 413,99 | -23.0% |

**Figure 45: Sensitivity Analysis - Comeback**

| Sensitivity Analysis | | | | | | |
|---|---|---|---|---|---|---|
| WACC | Growth Rate (g) | | | | | Sensitivity to Base Case |
| 11 635,66 | 2,0 % | 2,15% | 2,3 % | 2,45% | 2,5 % | |
| 7,0 % | 14 930,35 | 15 470,23 | 16 044,57 | 16 656,78 | 16 869,92 | +37.9% |
| 7,5 % | 12 731,21 | 13 157,86 | 13 609,13 | 14 087,20 | 14 252,93 | +17.0% |
| 8,0 % | 10 931,52 | 11 274,56 | 11 635,66 | 12 016,27 | 12 147,75 | 0.0% |
| 8,5 % | 9 437,68 | 9 717,52 | 10 010,90 | 10 318,82 | 10 424,89 | -14.0% |
| 9,0 % | 8 182,88 | 8 414,00 | 8 655,47 | 8 908,00 | 8 994,76 | -25.6% |

**Figure 46: Sensitivity Analysis - Grind**

| Sensitivity Analysis | | | | | | |
|---|---|---|---|---|---|---|
| WACC | Growth Rate (g) | | | | | Sensitivity to Base Case |
| ( 8 009,77) | 1,00% | 1,10% | 1,20% | 1,60% | 1,80% | |
| 7,0 % | ( 9 035,01) | ( 9 109,51) | ( 9 186,58) | ( 9 523,39) | ( 9 711,22) | -14.7% |
| 7,5 % | ( 8 430,35) | ( 8 491,14) | ( 8 553,86) | ( 8 826,00) | ( 8 976,40) | -6.8% |
| 8,0 % | ( 7 907,86) | ( 7 958,08) | ( 8 009,77) | ( 8 232,70) | ( 8 354,95) | 0.0% |
| 8,5 % | ( 7 451,27) | ( 7 493,19) | ( 7 536,26) | ( 7 721,03) | ( 7 821,69) | +5.9% |
| 9,0 % | ( 7 048,37) | ( 7 083,69) | ( 7 119,92) | ( 7 274,62) | ( 7 358,42) | +11.1% |

**Figure 47: Sensitivity Analysis - Fall**

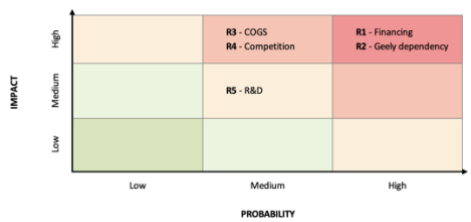



**Figure 48: Risk Matrix**

## COGS Structural Lock

Polestar needs volume to reduce COGS, but its COGS is so high that it cannot price competitively enough to generate volume. The Comeback scenario requires COGS to fall to 65% by 2035 a 78-percentage-point improvement through simultaneous economies of scale, battery cost declines, and contract renegotiations. COGS is the single variable with the largest valuation impact, with a $47B spread between best and worst case. We would revise our position if two consecutive quarters showed gross margin turning positive, evidencing that the cost inflection point has been structurally reached rather than temporarily achieved through mix effects.

## Chinese Competition

BYD's Han EV is now directly priced against the Polestar 2. NIO and XPeng are entering Europe, and Tesla's Model Q targets the same customer. These are present realities that already contributed to the revenue decline in 2024, and a sustained price war could force ASP reductions faster than COGS can be cut - the mechanism that drove Fisker to liquidation. We would revise our sell position if additional EU tariffs on Chinese EVs materially blunted BYD's and NIO's pricing advantage in Europe, or if Polestar demonstrated two consecutive years of volume growth above 30% without a corresponding decline in ASP, confirming resilient premium brand equity against competitive pressure.

## R&D Underinvestment

Polestar cut R&D from $175M (2022) to $38M (2024) - from 7% to 1.9% of revenue - while Tesla spends 3.5% and BYD over 5%. This preserves near-term cash but risks product obsolescence in the future. We would revise our recommendation if Polestar sustainably restored R&D spending above 4% of revenue for at least four consecutive quarters or secured a co-development agreement with Volvo or a third-party technology partner that credibly de-risks the Polestar 5 and 6 product timelines.

## CONCLUSION

Polestar presents an attractive brand and a differentiated positioning in the premium EV market. However, this positioning has not translated into economic profitability, as the company remains constrained by a structurally unprofitable cost base and limited scale. In addition, its highly leveraged balance sheet and dependence on external financing significantly increase execution risk. While upside exists in a successful turnaround scenario, it requires sustained and uncertain improvements across multiple drivers.

Overall, we believe the risk-reward profile remains unfavorable and reiterate our **SELL** recommendation.

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Disclaimer: An AI tool (ChatGPT) was used to assist with the stylistic refinement of this text.

# Annex

## Annex 1 – Beta [Team analysis]

| Parameter | Optimistic | Pessimistic | Average (used in model) |
|---|---|---|---|
| Risk-Free Rate (rf) | 4,00% | 4,00% | 4,00% |
| Market Risk Premium (MRP) | 5,00% | 6,00% | 5,50% |
| Asset Beta | | | |
| Industry Beta - Automotive sector unlevered beta (Damodaran) | 0,80 | 0,80 | 0,80 |
| Observed cost of debt kD (Interest / Avg. Debt, 2024) | 7,31% | 7,31% | 7,31% |
| Credit spread over rf (kD – rf) | 3,31% | 3,31% | 3,31% |
| βD (Schaefer & Strebulaev, B-rated debt) | 0,26 | 0,26 | 0,26 |
| Implied βD from CAPM [(kD – rf) / MRP] — for reference | 0,66 | 0,55 | 0,60 |
| kD via CAPM: rf + βD × MRP | 5,30% | 5,56% | 5,43% |
| kD observed (market rate on debt) | 7,31% | 7,31% | 7,31% |
| Levered βE = βA + D/E × (βA – βD) [Hamada formula] | Not applicable — negative equity (D/E undefined) | | |
| kE via asset-level identity: kA + D/E × (kA – kD) | Not applicable | | |
| kA = rf + βA × MRP [CAPM applied at asset level] | 8,00% | 8,80% | 8,40% |
| WACC used in model (= kA, since equity is negative) | 8,00% | 8,00% | 8,00% |
| Sensitivity range for kA (βA = 0.70 to 1.00) | 7,50% | 10,00% | |

## Annex 2 – Beta explications [Team analysis]

| Explication |
|---|
| US 10-Year Treasury yield (proxy for long-run risk-free rate). Swedish/EUR equivalent (Riksbank) broadly consistent at ~3.8-4.2%. Per Block 4 slides: 'preferably a long-run zero-coupon government bond'. |
| Range 5–6% per course framework. Consistent with long-run historical estimates: Dimson-Marsh-Staunton global MRP ~5.0-5.5%; Damodaran US implied ~4.5-5.5%. 5% = optimistic (lower risk perception); 6% = pessimistic (higher uncertainty). |
| Asset Beta - Polestar's book equity is NEGATIVE (–$3.33B in 2024) and its D/E ratio is therefore economically meaningless. The traditional relevering formula kE = kA + D/E × (kA – kD) breaks down. We therefore work at the ASSET (unlevered) level — βA captures the risk of the operating business independently of capital structure, which is the appropriate input to the unlevered WACC (= kA) used to discount NOPAT in our DCF. An EV 'startup' beta (e.g. Tesla raw equity beta ~1.8-2.0) would overstate risk for valuation purposes: (1) Polestar's revenues are 100% from car sales — not software or services; (2) its cost structure (COGS 143% of revenue) is industrial, not asset-light; (3) it operates in a mature regulated market. The automotive unlevered beta (0.80) is our central estimate. A sensitivity range of 0.70–1.00 is shown in the Sensitivity Analysis sheet. |
| Damodaran (Jan 2024) average unlevered beta for Global Auto & Truck sector: ~0.75-0.85. We use 0.80 — treating Polestar as an automotive company (revenue from car sales, manufacturing cost structure). This is consistent with the 8% WACC used in our model: 8% = 4% + 0.80 × 5% (MRP optimistic). |
| Interest expense 2024: $314.3M / Average debt ($3,587.7M + $5,012.4M)/2 = $4,300.1M → kD = 7.31%. Source: Reorganised Income Statement & Balance Sheet. |
| Spread of ~330 bps corresponds to approximately B/B+ credit rating range (per Block 4 slide, Schaefer & Strebulaev table). Polestar: negative equity, 182% debt/assets → sub-investment grade confirmed. |
| Per Block 4 slide Table 12.3: B-rated debt avg. beta = 0.26. βD ≠ 0 because kD = 7.31% >> rf = 4.0%. Setting βD = 0 would imply risk-free debt — inconsistent with Polestar's distressed credit profile. Note: implied CAPM beta from kD would be ~0.66 (MRP=5%), but we prefer the empirical table value which captures only systematic (non-diversifiable) default risk. |
| Implied CAPM beta from observed kD. High value (~0.60-0.66) includes idiosyncratic default premium, not just systematic risk — hence we prefer the Schaefer table (C23). Both confirm βD >> 0. |
| CAPM-implied kD using systematic beta only. Lower than observed kD (7.31%) because the CAPM strips out idiosyncratic default risk. Used for theoretical consistency. |
| Preferred for practical WACC calculation. Reflects actual financing cost. The gap vs CAPM kD = default risk premium (idiosyncratic) embedded in Polestar's bonds. |
| Polestar's book equity is NEGATIVE. The traditional relevering formula cannot be applied mechanically. We derive kE for informational purposes using the unlevering identity, but the result is not economically meaningful for a firm with negative equity. The operative discount rate in our DCF is kA (= WACC unlevered).. D/E = debt / equity = 5,012 / (-3,329) < 0. The Hamada formula yields a negative levered beta, which is economically meaningless. |
| Slide formula: kE = kA + D/E × (kA – kD). Breaks down with negative equity. For Polestar: all equity value is a real option (upside if Comeback materialises). Market cap ($1.5B) << Enterprise Value ($6.7B) — option-like pricing. |
| Central formula from Block 4: ri = rf + βi × (rm – rf). Applied at asset (unlevered) level. Optimistic: 4% + 0.80×5% = 8.0% ✓ Consistent with WACC=8% used throughout our model. |
| With negative equity the traditional WACC = E/(D+E)×kE + D/(D+E)×kD×(1-T) is not directly applicable. We use the unlevered cost of capital kA = 8% as the discount rate — consistent with discounting NOPAT to value operating assets (VOA). Tax rate = 0% (company has never been profitable; no tax shield to capture). |
| Lower bound: βA=0.70, MRP=5% → kA=7.5%. Upper bound: βA=1.00, MRP=6% → kA=10.0%. See Sensitivity Analysis sheet. Base case 8% sits comfortably in the middle of this range. |